
How America Talks: Quantifying and Predicting Dialect Variation Across the Continental United States

First Author

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Abstract

This is where you will write an abstract of about 150-250 words. The abstract should (1) briefly introduce the topic and why it's important, (2) describe your dataset; (3) describe what you did; (4) describe what you found; (5) say why your results are important or useful.

1 Introduction

Our project explores how English dialects vary across the United States, curious about how people communicate the same ideas using different regional vocabularies and grammatical patterns. Using a large dialect dataset originally collected by Professor Bert Vaux, we cleaned user, question, choice, and response CSV files to prepare them for analysis. His dataset contained hundreds of thousands of survey participants, over 160 questions, thousands of answer choices, and more than 11 million responses. We focused on key demographic and linguistic fields, removing duplicates, addressing extensive missing data, and standardizing question and choice formats.

With this cleaned dataset, we conducted both clustering and predictive modeling. K-Means clustering showed very low alignment with true geographic regions, indicating that dialect-based patterns do not naturally recreate formal regional boundaries, showing our poor clustering results. Logistic regression predictions performed moderately, with precision, recall, and F1-scores varying by region, with an overall accuracy around 0.57 and the highest being XGBoost with an accuracy of 0.59, which is what we used for the regional dialect prediction quiz. We also built an interactive dashboard that summarizes the dataset, provides visualizations, produces graphs, and includes the regional dialect prediction quiz, allowing users to see which region their own dialect most resembles.

Overall, our work highlights how language reflects culture, history, and identity, helping people appreciate everyday linguistic diversity. These findings can support educators, researchers, and organizations by revealing how dialects shift across regions and by offering insights for communication, cultural analysis, and future linguistic studies.

2 Prior Work

2.1 What should go into this section

Way back in the beginning of the semester, you each reviewed a paper. You will include those reviews here, but you'll need to write some text to connect them to each other, edit them for length, and make sure you connect that work clearly to the work you did.

You will need to **add 4 additional sources**. These can be about specific methodologies, unlike the first lit review you did where I wanted you to focus on the actual topic of your project. You still should not be citing textbooks, though you could potentially cite a chapter of a textbook. In total, you should have 8 (or 9 for the marathon team) papers, articles, and book chapters (but not books!) reviewed here.

If my feedback on your lit review article way back at the beginning said "You need to find a different article for your term paper" you will need to find a different article for your term paper. That means

you personally might have to contribute 2 new reviews – one to make up for the one you did last time that had a problem and a new one.

Please take a look at the prior work sections of articles and papers to understand the structure and style of a literature review. (Feel free to look at any of the papers on my website.) It shouldn't be just a disconnected series of paragraphs, each one about a different paper. You need just a sentence or two about each source, and you need to connect them to each other and to your paper in a coherent way.

***Research on regional and social variation in American English demonstrates that linguistic forms—whether lexical, grammatical, or phonetic—often signal aspects of identity, though not always in reliable or uniform ways. Axelrod and Scheibman (2013) provide a wide-ranging descriptive overview of contemporary U.S. English, surveying sociolinguistic research on features such as y'all, fixin' to, double modals, vowel shifts, quotatives, and ethnic varieties like African American English or Chicano English. Their work emphasizes that while such forms can index region, age, ethnicity, or gender, their social meanings are fluid and context-dependent. This broad descriptive framework establishes the kinds of linguistic variables that might differentiate regional varieties and informs our own focus on which words most consistently function as regional cues.

Quantitative studies of lexical variation further clarify how vocabulary can map dialect boundaries. Boberg's (2005) North American Regional Vocabulary Survey demonstrates that lexical choices can reliably distinguish Canadian regions from one another and from the United States. By calculating net variation across 53 lexical variables and identifying major isoglosses—such as the contrast between cabin and cottage—Boberg shows how aggregated lexical decisions reveal large-scale regional divisions. His methodological approach is directly relevant to our project: like his survey, our region-prediction quiz treats lexical items as measurable variables capable of differentiating speakers based on region.

While these studies focus on production, research in dialect perception raises important questions about the reliability of identifying social or regional categories from linguistic features alone. Lee's (1971) critical examination of early dialect-perception work found low interrater and intrarater reliability when listeners judged social attributes based solely on speech samples. His findings suggest that human judgments of dialect can be inconsistent, shaped by expectations about race or class rather than stable linguistic cues. This serves as a caution for any task involving dialect classification, including ours: without carefully selected and standardized stimuli, perceived dialect differences may reflect noise or stereotype rather than genuine linguistic structure.

A more recent perceptual study by Clopper, Levi, and Pisoni (2006) offers a systematic account of how listeners categorize American regional dialects. Using speech samples from four regional groups and a combination of forced-choice and paired-comparison tasks, they found modest accuracy in direct classification but stronger and more consistent patterns in perceived similarity. Their multidimensional scaling results reveal two major perceptual dimensions—geographic proximity and dialect markedness—indicating that listeners rely on broad perceptual groupings rather than fine-grained linguistic distinctions. These findings parallel the challenges we observed in our quiz: while participants can sometimes identify a speaker's region from lexical choices, accuracy varies, and judgments often cluster into broad regional categories.

These works establish both the promise and the limitations of using linguistic features to infer regional identity. Axelrod and Scheibman outline the types of forms that can serve as markers; Boberg demonstrates how lexical variables can quantify regional boundaries; and perceptual research shows that classification of dialects—by humans or models—is inherently imperfect. Our region-prediction quiz builds on these insights by testing, through an interactive lexical-choice task, how reliably contemporary speakers perceive regional identity and which words provide the strongest cues. The mixed performance we observed aligns with the broader literature: regional variation is real, but its perceptual salience varies, and listeners often rely on only a subset of salient lexical markers.

2.2 How to do citations with this template

Put your articles and papers in the bibliography.bib file in this project using the BibText format. I have given you one example. You can find out more about BibText entries and how they should be formatted here: <https://www.bibtex.com/e/entry-types/>

Once you’ve put the article you want to cite in the .bib file, you can then cite it in the text. You have two commands you can use:

- `\citep`
- `\citet`

The `\citep` command will produce the authors and year in parentheses, like this:

Prior work has found an association between online and offline gang conflict (Leverso et al., 2025).

The `\citet` command will produce the authors’ names outside parentheses (so you can use them grammatically in a sentence) and then the year in parentheses:

Leverso et al. (2025) found an association between online and online gang conflict.

3 Data

The dataset originates from Dr. Bert Vaux, Professor of Phonology and Morphology at the University of Cambridge, who conducted dialect surveys. He began collecting them on paper before later migrating them to an online format with assistance from computer science students. Users in the survey were asked a question and had to choose one choice for their response (Figure 1). Dr. Emily Prud’hommeaux, Professor of Computer Science at Boston College, subsequently organized the data into a regional dataset containing four CSV files (Table 1).

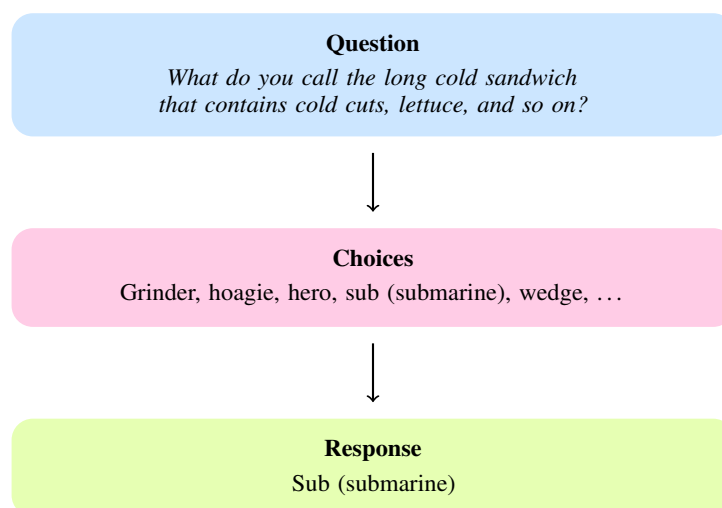


Figure 1: Example of a dialect-survey question, available choices, and an individual user’s response.

CSV File	Samples (rows)	Features (cols)
questions	181	7
choices	1484	6
users	385780	20
responses	11853462	7

Table 1: Dataset overview: number of samples and features in each CSV file prior to preprocessing.

Each CSV file contains identification fields stored as integers to link samples between files, text-based fields for objects like name and open-ended responses, boolean indicators encoded as integers, and float fields for geographic information.

Prior to analysis, we performed several preprocessing steps. First, we deduplicated survey questions that appeared in multiple survey versions by mapping shared question IDs across *questions*, *choices*, and *responses*. Additionally, we restricted the dataset to users whose reported latitude and longitude fell within the continental United States.

We restricted our analysis to questions that received between 40,000 and 60,000 responses. A histogram of response counts (Figure 2) showed that this range represented the majority of questions in the dataset, allowing us to retain the most consistently answered items while removing outliers with unusually low or high response rates. The full list of the 108 retained survey questions can be found in **Appendix A**.

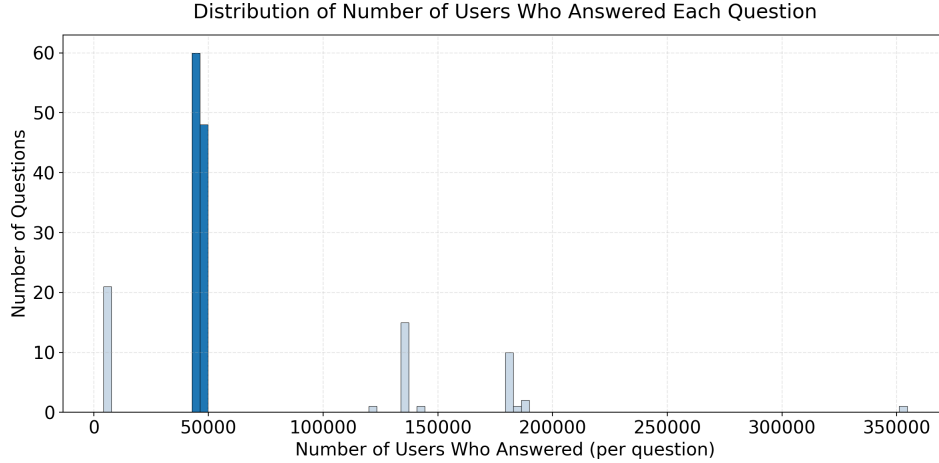


Figure 2: Histogram of User Response Counts per Question, Highlighting the 40–60k Interval

Ground-truth regions were assigned to each user based on the defined dialect regions from the independent work of Rick Aschmann, who drew directly from The Atlas of North American English. Distribution of users across dialect regions is shown in Figure 3.

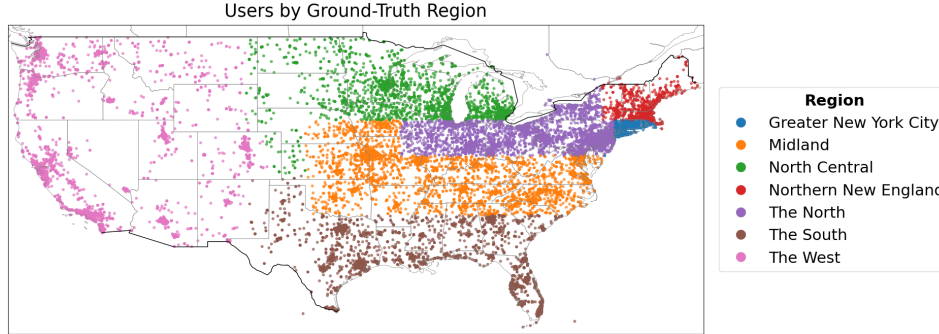


Figure 3: Post-Processing Geographic Distribution of Users by Ground-Truth Dialect Region

Statistic	Value
Unique Questions	108
Unique Choices	557
Total Responses	4,973,285
Responses per Question (min–max)	43,387 – 47,034
Responses perxg Question (mean)	46,048.9
Unique Users in Final Dataset	47,218

Table 2: Dataset statistics after preprocessing and filtering.

After merging the four datasets and applying all preprocessing steps, the final dataset contains nearly five million responses to more than 100 questions from over 47,000 unique users (Table 2).

3.1 Figures and Tables

Here's some information about Figures and Tables relevant for this template. I am including these instructions here in the Data section, but these instructions apply throughout the paper. I will take off points for violating these instructions.

- Every figure or table **must have a caption** explaining how to interpret that figure or table. See Table 3 and Figure 4 for examples.
- Remember that latex numbers tables and figures for you! You do not need to keep track of the tables or numbers yourself.
- In case you missed it the first thousand times I said this, figures must be legible. As I said a lot when giving you feedback on your posters, make the text bigger first, then make the visualization bigger.
- Make your tables natively in latex. Latex tables are challenging, so I recommend making your tables **with a nice latex table generator tool online** (e.g., <https://www.tablesgenerator.com/>). Do not screenshot tables you made elsewhere unless it is truly impossible to create your table in latex.
- If you have images that you did not create in a figure, be sure to provide the proper attribution.
- **Every table and figure must be referenced in the text somewhere.**

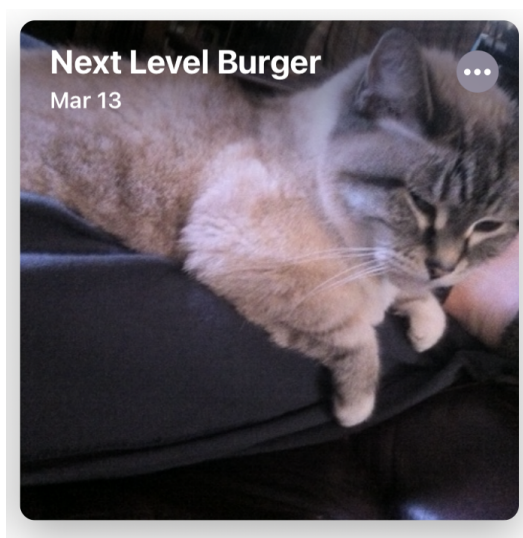


Figure 4: Scientists are still trying to understand why the Photos app on my iPhone created a compilation of photos of my old cat entitled “Next Level Burger”. Please note that the cat does not resemble a burger, which suggests that Apple’s AI needs some improvement.

Year	Country	Artist	Song
2025	Austria	JJ	Wasted Love
2024	Switzerland	Nemo	The Code
2023	Sweden	Loreen	Tattoo
2022	Ukraine	Kalush Orchestra	Stefania
2021	Italy	Maneskin	Zitti e Buoni

Table 3: Results from the last five years of the Eurovision Song Contest. We see that songs with English language titles are more common. In addition, countries where Germanic languages are spoken (German and Swedish) are more frequent winners.

4 Statistical Analysis

If your work included statistical analysis, describe your method and results here. Its better to provide more detail. You should tell me:

- What techniques did you use? Provide a good description of how that technique works.
- For our statistical analysis we employed the chi-square test of independence to determine whether age groups and dialect term usage are statistically associated. This test evaluates whether the observed distribution of categorical variables differs from what can be expected if the two were completely independent. The chi-square test works by first creating a contingency table showing the combination. For example how many people born within a certain age range respond with "roly poly" vs "pillbug." Then, it calculates the expected frequencies for each example to if the example was independent, which assumes that age and word choice are unrelated. The test computes the chi-square statistic using the formula $X^2 = \sum \left[\frac{(Observed - Expected)^2}{Expected} \right]$ summed across all the examples in the contingency table. The chi-square value is then compared against a theoretical chi-square distribution to determine the p-value, which represents the probability of getting the observed data in relation to the assumption that there is no relationship. So, a lower p-value score ($p < 0.05$) indicates statistical significant association meaning we can conclude that age group and dialect choice are not independent. Following the chi-square, we calculated Cramér's V as a standardize effect size measure. Cramér's V provides a measure of association strength ranging from 0 (no association) to 1 (perfect association). This normalization allows meaningful comparison of effect sizes across different questions with different numbers of response options and age groups. We interpreted Cramér's V using these thresholds: negligible ($V < 0.10$), small effect ($0.10 \leq V < 0.30$), medium effect ($0.30 \leq V < 0.50$), and large effect ($V \geq 0.50$). Doing this approach with chi-square and Cramér's V identifies differences that both statistically reliable and meaningful for understanding generational dialect usage.
- What, if anything, did you do with your data (e.g., perhaps you only used some columns, fields, or samples)?
- The changes we did to our data were mainly text normalization. For example, answers that had a variety of spellings were set to the most common form. For example, "roly poly", "rolly polly", and "rollie pollie" was changed to just "roly poly". Furthermore, some answers were just very long and were shortened to have the same meaning. Lastly, some answered shared one answer but had another variant, like "sub" and "sub (submarine)" were merged into just "sub (submarine)". Then for our geographic analysis, the dataset was filtered to include only United States respondents and were classified into one of the nine US dialect regions based on their latitude and longitude coordinates. This regional dataset was merged back into the main dataset, and responses with missing geographic information were excluded. For our temporal analysis, the dataset was just split into two birth groups: users born before 1975 and those born after 1975.
- How did you evaluate or measure the results?
- We evaluated our results using the techniques mentioned, such as chi-square test, p-value, and Cramér's V score. We used the top 5 questions that had the most responses to ensure that our results was backed up with enough data. When looking at these 5 questions which were "What is a long cold sandwich with cold cuts and so on?", "What do you call the kind of rain when the sun is shining?", "What do you call gooey or dry matter that collects in the corners of your eyes?", "What do you call a small creature that rolls into a ball when you touch it", and "How do you pronounce the vowel sound in the word aunt?" For all of these top 5 questions, we found that they had a very large value for the chi-square, meaning that time and age is a real pattern. Then by looking at the p-value score, we can see that
- Provide an interpretation of the results.
- Include graphs or tables showing your results.

5 Machine Learning

To prepare the survey data for modeling, we transformed all user responses into a high-dimensional one-hot encoded matrix aligned across all users and all possible answer choices. Each feature corresponds to a unique (question, answer) pair drawn from the preprocessed dataset. If a user did not answer a particular question, all features corresponding to that question’s answer choices were set to 0. This creates an artificial "missing" category and keeps the dimensional structure fixed, which is necessary for linear models. The dataset contains 47,128 users, each represented by 108 questions which were encoded into 665 question–choice features for each user. An example of this is shown in Table 4. Two machine learning analyses were conducted using this dataset with different task-specific preprocessing steps applied.

User ID	243_MISSING	243_[o:] as in “Bo”	243_[u:] as in “boo”	Ground-Truth Region
1126637	1	0	0	West
1126639	0	1	0	Northern New England

Table 4: Example of one-hot encoded inputs for Question 243 (*the first vowel in "Bowie knife"*) showing a subset of response categories. Ground-truth region labels are excluded from training and used only for evaluation; user IDs are likewise omitted from training and retained only for record-keeping.

5.1 Region Prediction Quiz

Inspired by the *New York Times* region-prediction quiz (CITE HERE!!!), we aimed to create a shorter and more effective quiz using ML techniques. Rather than relying on the full set of 108 questions, our goal was to determine whether a small group of especially diagnostic lexical variables could approximate a user’s dialect region with reasonable accuracy.

We trained and tested five classification models on the question-answer choice features, sampled in Table 4. The data was split into 80% for training and 20% for testing. Classifier models and their hyperparameters are shown in Table 5.

Model	Hyperparameters
Logistic Regression	multinomial; lbfgs; max_iter=1000
Random Forest	300 trees; unlimited depth; n_jobs=-1; random_state=42
Gradient Boosting	300 estimators; learning rate = 0.05; max depth = 3; subsample = 0.8
Linear SVM	C=1.0; class weight = balanced; max_iter=5000
XGBoost	300 trees; learning rate = 0.05; max depth = 6; subsample = 0.8; colsample_bytree = 0.8; objective = multi:softprob

Table 5: Supervised learning models and their hyperparameters used for dialect region classification.

Majority and random baselines were also used for comparison. The majority baseline predicts the most common region for all users (The North), which provides a benchmark for class imbalance. The random baseline assigns regions by sampling from the training-set class distribution, simulating a random guess. Results are shown in Table 6.

All models achieved higher accuracy than both baselines, but XGBoost performed the strongest, with an accuracy of 0.59. It also maintained the strongest precision (0.61) and recall (0.56) across regions relative to the other models. Table 7 shows detailed performance metrics for XGBoost across regions.

Once XGBoost was determined to be the strongest model, SHAP (Shapley Additive Explanations) was used to find prominent lexical variables for each region, as determined by XGBoost. Results are shown in Table 8. SHAP assigns each feature a value indicating how much it contributes to a model’s prediction. For a model prediction $f(x)$ and feature i , the SHAP value is shown below:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(x_{S \cup \{i\}}) - f(x_S)],$$

Model	Precision	Recall	F1-Score	Accuracy
<i>Random Baseline</i>	–	–	–	0.17
<i>Majority Baseline</i>	–	–	–	0.24
Random Forest	0.54	0.49	0.51	0.53
Logistic Regression	0.57	0.54	0.55	0.57
Gradient Boosting	0.59	0.55	0.57	0.58
Linear SVM	0.53	0.56	0.54	0.55
XGBoost	0.61	0.56	0.58	0.59

Table 6: Macro-averaged supervised classification performance, with random and majority baselines for reference.

Region	Precision	Recall	F1-Score
Greater New York City	0.58	0.45	0.51
Midland	0.57	0.59	0.58
North Central	0.66	0.59	0.63
Northern New England	0.70	0.51	0.59
The North	0.53	0.65	0.58
The South	0.54	0.46	0.50
The West	0.65	0.66	0.66
Overall Accuracy	0.59		

Table 7: Per-region performance of the XGBoost classifier.

F is the full set of features, S is any subset of features not including i , $f(x_S)$ is the model prediction using only the features in S , and ϕ_i is the SHAP value quantifying feature i ’s contribution.

Region	Most Predictive Feature	Mean SHAP
Greater New York City	Q255: pronouncing Mary, merry, and marry all differently (Mary/merry/marry)	0.2370
Midland	Q305: calling the glowing summer insect a “firefly”	0.2076
North Central	Q316: calling something diagonally across the street “kitty-corner”	0.2828
Northern New England	Q266: pronouncing “route” to rhyme with “hoot”	0.1256
The North	Q305: calling the glowing summer insect a “firefly”	0.1669
The South	Q343: calling the school drinking fixture a “water fountain”	0.1448
The West	Q319: using “highway” as the general term for a big, fast road	0.2938

Table 8: Top region-distinguishing lexical variables according to mean absolute SHAP value.

5.2 Clustering

In addition to supervised learning, unsupervised learning was also conducted to determine if the features would naturally cluster by region. K-Means with Euclidean distance was selected as the clustering algorithm because it is well-suited for high-dimensional and sparse data like our one-hot encoded survey matrix. The one-hot encoded features (Table 4) were scaled using Z-score scaling for clustering so that each answer category contributed equally to Euclidean-distance calculations.

The K-Means algorithm partitions the dataset into k clusters by minimizing the total within-cluster variance. Its objective function is shown below, where μ_j is the centroid of cluster C_j :

$$\min_{\{C_1, \dots, C_k\}} \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2,$$

K-Means clustering was evaluated for $k \in \{3, 4, 5, 6, 7\}$. Two different feature representations were used: (1) the full scaled one-hot encoded response matrix and (2) a reduced set of the top 50 most region-predictive features from Logistic Regression in section 5.1 (see **Appendix B** for top-50

question list). Performance was assessed using the silhouette score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI).

The silhouette score measures within-cluster cohesion relative to between-cluster separation. Values close to 1 indicate well-separated clusters, while values near 0 indicate overlapping clusters. Negative values imply that points may be assigned to the wrong cluster. It is defined below, where $a(i)$ is the average intra-cluster distance and $b(i)$ is the nearest-cluster distance:

$$\text{Silhouette}(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

The ARI evaluates agreement between cluster assignments and ground-truth dialect regions, correcting for chance. ARI ranges from 0 (random labeling) to 1 (perfect recovery), with negative values indicating worse-than-chance alignment. It is defined below, where RI is Rand Index (unadjusted):

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]},$$

The NMI measures shared information between the clusters and true labels. NMI ranges from 0 (no shared information) to 1 (perfect correspondence). It is defined below, where C is the predicted cluster assignment, R is the ground-truth region label, $I(C; R)$ is the mutual information between I and R , and $H(\cdot)$ denotes entropy:

$$\text{NMI}(C, R) = \frac{2 I(C; R)}{H(C) + H(R)}.$$

k	Silhouette (Full)	ARI (Full)	NMI (Full)	Silhouette (Top)	ARI (Top)	NMI (Top)
3	0.0235	0.0228	0.0377	0.0504	0.0373	0.0574
4	0.0233	0.0190	0.0347	0.0185	0.0931	0.1251
5	0.0117	0.0736	0.1009	0.0274	0.0907	0.1207
6	0.0130	0.0543	0.0823	-0.0399	0.0841	0.1159
7	0.0162	0.0711	0.1133	0.0140	0.0821	0.1155

Table 9: Comparison of K-Means clustering performance on the full feature set versus the top region-predictive features, evaluated using Silhouette, ARI, and NMI.

Across all values of k , both the full feature set and top-feature subset produce very weak clustering structure. The Silhouette score ranges from approximately -0.04 to 0.05, which indicates that there is no meaningful geometric cluster structure among the encoded questions and answers. Similarly, the ARI ranges from approximately 0.02 to 0.09, indicating that the performance was equivalent to random assignment. Lastly, NMI ranged from 0.03 to 0.11 for the full feature clustering and ranged from 0.05 to 0.13 for the top-feature clustering. Although these values indicate that clusters share minimal information with the true dialect regions, the top-feature subset shows slightly higher NMI.

The strongest clustering solution was $k = 5$ using the top-question feature subset. It included one of the strongest ARI (0.09) and NMI (0.12) values, and maintained a competitive silhouette score (0.027) relative to other k -values. The number of users in each region in these clusters is shown in Table 9, and a visual distribution of cluster users is shown in Figure 5. Clusters 1 and 3 are the largest and dominate the map, with Cluster 1 dominating the North Central and West regions and Cluster 3 dominating the Midland, South, North, Greater New York City, and Northern New England regions.

Region	C0	C1	C2	C3	C4	Total Users
Greater New York City	53	442	125	384	2,547	3,551
Midland	118	2,530	176	5,328	845	8,997
North Central	90	4,674	198	1,100	526	6,588
Northern New England	43	458	103	378	1528	2,510
The North	173	3,913	309	3,963	3,136	11,494
The South	458	1,138	154	3,186	637	5,573
The West	120	6,037	253	1,405	690	8,505
Cluster Totals	1,055	19,192	1,318	15,744	9,909	47,218

Table 10: Region-by-cluster distribution for K-Means with $k = 5$ using the top region-predictive features. The final column reports total users per region (not used in clustering); the bottom row shows total users per cluster.

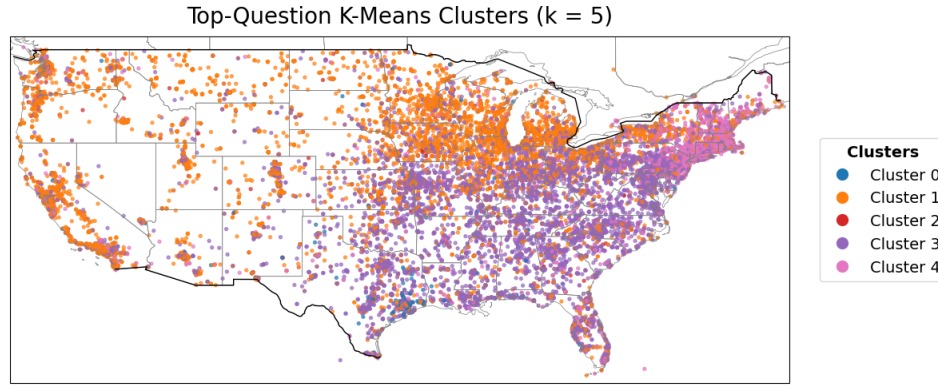


Figure 5: Geographic distribution of K-Means clusters for $k = 5$ using the top region-predictive features. Colors indicate cluster users within each dialect region.

If your work included machine learning, describe your method and results here. Its better to provide more detail.

- What techniques and algorithms you used. You don't need to explain in detail how this algorithm works, but a very high-level description is good (e.g., is it neural? is it linear? is it generative or discriminative?).
- What, if anything, did you gave to do with your data (e.g., perhaps you only used some columns, fields, or samples, maybe you created a one-hot vector)?
- What parameterizations or hyperparameterizations did you select?
- What are your inputs and output (e.g., how many features, how many possible classes, what are the distributions of the classes)?
- How did you partition your data or do cross validation? Provide specific info like "X training samples, Y testing samples".
- You must provide at least one baseline, ideally more than one!
- Include graphs or tables showing your results. Remember that for classification you should never show only accuracy! Overall accuracy (i.e., percent correct) is not particularly meaningful. You need to show other eval metrics, like precision, accuracy, ROC, etc.

6 Data Communication

To communicate our findings clearly and give users an interactive way to explore American dialect patterns, we created a Streamlit dashboard that summarizes our dataset, visualizes key trends, and includes our own version of the NYT dialect quiz, allowing users to see which region their speech most closely aligns with. See Figure 6, 7, 8, and 9 for pictures from the dashboard.

The dashboard helps users living in the United States understand how language reflects culture, history, and identity while highlighting familiar variations in everyday speech, like “soda” vs. “pop” vs. “coke.” Built using VS Code, Python packages like Streamlit, Pandas, numpy, plotly, gdown, re, pathlib, GitHub for version control, Google Drive for hosting data, and deployed on Streamlit Cloud, the dashboard follows a simple architecture where our cleaned datasets are fed into visualizations and prediction tools accessible through an intuitive user interface. This tool benefits students, educators, researchers, and organizations as it promotes linguistic awareness, supports studies of dialect change and migration, and helps groups adapt their communication to regional audiences.

Through developing the dashboard, we gained a deeper understanding of how dialects vary across the country and how data-driven tools can make this linguistic diversity more accessible. The dashboard will be demonstrated during our poster presentation, and it is publicly available here (must be in light mode), with the full codebase on Github.

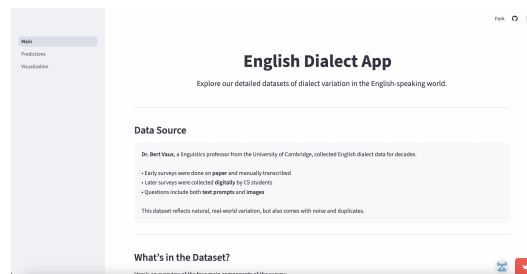


Figure 6: Screenshot of the Main Page of the dashboard.

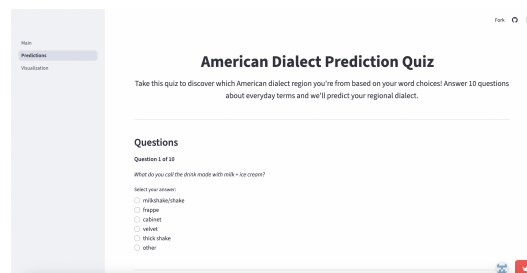


Figure 7: Screenshot of the Prediction Page of the dashboard.

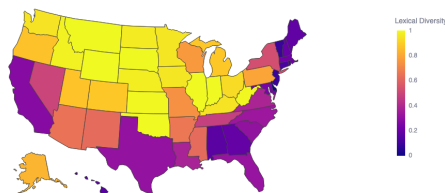


Figure 8: Screenshot of an example of a visualization on the Visualization Page of the dashboard.

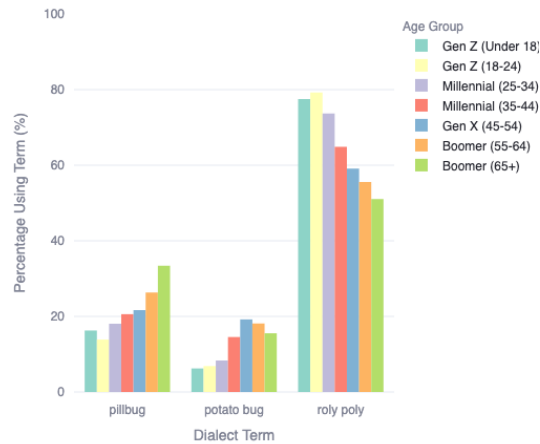


Figure 9: Screenshot of the bar graph of usage by age group on the Visualization Page of the dashboard.

7 Conclusions and Future Work

Make sure to cover all of these:

- Summarize your results.
- Explain why they are important or interesting.
- What can other people learn or gain from your project.
- What made your project difficult and how did you overcome those difficulties?
- **Three different things** you would you like to explore if you could continue this project.

8 Appendix

Appendix A: Survey Questions

Number of questions: 108

- Question 242: *been*
 - as in "sit"
 - i as in "see"
 - as in "set"
 - other
- Question 243: *the first vowel in "Bowie knife"*
 - o as in "Bo"
 - u as in "boo"
 - I have seen this word in print, but have no idea how to pronounce it
 - I have never seen or heard this word
 - other
- Question 244: *caramel*
 - with 2 syllables ("car-ml")
 - with 3 syllables ("carra-mel")
 - I use both interchangeably

- (d) I have both forms, but the two have different meanings (please state how in the comments box)
 - (e) other
4. Question 245: *the vowel in the second syllable of "cauliflower"*
- (a) i as in "see"
 - (b) as in "sit"
 - (c) other
5. Question 246: *the last vowel in "centaur"*
- (a) as in "car" ("sen-tar")
 - (b) as in "caught"
 - (c) I use the same vowel in "car", "caught", and "centaur"
 - (d) rhymes with "sore" and "more" ("sen-tore")
 - (e) other
6. Question 247: *coupon*
- (a) with u as in "coop" ("coo-pon")
 - (b) with ju as in "cute" ("cyoo-pon")
 - (c) other
7. Question 248: *Craig (the name)*
- (a) as in "set"
 - (b) e as in "say"
 - (c) I say something in between the vowels in "set" and "say", but closer to the one in "say"
 - (d) I say something in between the vowels in "set" and "say", but closer to the one in "set"
 - (e) other
8. Question 249: *crayon*
- (a) as in "man" (1 syllable, "cran")
 - (b) ej (2 syllables, "cray-ahn")
 - (c) ej (2 syllables, "cray-awn", where the second syllable rhymes with "dawn")
 - (d) aw (I pronounce this the same as "crown")
 - (e) other
9. Question 250: *creek (a small body of running water)*
- (a) i as in "see"
 - (b) as in "sit"
 - (c) I use both interchangeably
 - (d) I don't know how to pronounce this word
 - (e) I use both, but they mean two different things (please state how they differ in the comments box)
 - (f) other
10. Question 251: *the first vowel in "Florida"*
- (a) o as in "flow" ("flow-ri-da")
 - (b) as in "ah" ("flah-ri-da")
 - (c) as in "saw" ("flaw-ri-da")
 - (d) as in "sore" ("flore-i-da")
 - (e) other
11. Question 252: *flourish*
- (a) as in "bird" ("flurr-ish")
 - (b) as in "sore" ("flore-ish")
 - (c) as in "sun" ("fluh-rish")
 - (d) other (including if you use one pronunciation for the verb and a different pronunciation for the noun)

12. Question 253: *the last vowel in "handkerchief"*
 - (a) i as in "see"
 - (b) as in "sit"
 - (c) other
13. Question 254: *lawyer*
 - (a) with j as in "boy" ("loyer")
 - (b) with as in "saw" ("law-yer")
 - (c) I use both interchangeably
 - (d) other
14. Question 255: *How do you pronounce Mary/merry/marry?*
 - (a) all 3 are the same
 - (b) all 3 are different
 - (c) Mary and merry are the same marry is different
 - (d) merry and marry are the same Mary is different
 - (e) Mary and marry are the same merry is different
15. Question 256: *mayonnaise*
 - (a) with as in "man" (2 syllables—"man-aze")
 - (b) with ej (3 syllables—"may-uh-naze")
 - (c) I use both interchangeably
 - (d) other
16. Question 257: *the first vowel in "miracle"*
 - (a) i as in "near"
 - (b) as in "knit"
 - (c) as in "net"
 - (d) I say something in between and
 - (e) other
17. Question 258: *mischievous vs. mischievious*
 - (a) mischievous (3 syllables)
 - (b) mischievious (4 syllables)
 - (c) I write "mischievous" but say "mischievious"
 - (d) I use both
 - (e) other
18. Question 259: *the final vowel in "Monday," "Friday," etc.*
 - (a) e as in "say"
 - (b) i as in "see"
 - (c) I use e with the words in isolation, but i in compounds (such as "Sunday school")
 - (d) other (e.g. do you use one vowel in some day names, and another in the other names?)
19. Question 260: *the second vowel in "pajamas"*
 - (a) as in "jam"
 - (b) as in "father"
 - (c) other
20. Question 261: *pecan*
 - (a) pikn with stress on the first syllable ("PEE-can")
 - (b) pikn with stress on the second syllable ("pee-CAN")
 - (c) pikn with stress on the first syllable ("PEE-Kahn")
 - (d) pikn with stress on the second syllable ("pee-KAHN")
 - (e) pkn ("pick Ann")
 - (f) pkn ("pick Ahn")

- (g) I pronounce it differently when it's alone than when it's in a compound like "pecan pie" (please state how you pronounce the two variants in the comments box)
 - (h) other
21. Question 262: *poem*
- (a) one syllable
 - (b) two syllables
22. Question 263: *really*
- (a) i as in "see" ("reely")
 - (b) as in "sit" ("rilly")
 - (c) i ("ree-l-y")
 - (d) other (including if you use two or more of these interchangeably)
23. Question 264: *realtor (a real estate agent)*
- (a) syllables ("reel-ter")
 - (b) syllables (realtor, in other words "reel-uh-ter")
 - (c) syllables (ree-l-ter)
 - (d) I don't use this word I use "estate agent"
 - (e) other
24. Question 265: *roof, room, broom, root*
- (a) u as in "food"
 - (b) as in "foot"
 - (c) these four words do not all have the same vowel (please use the comments box to let us know which is which)
25. Question 266: *route (as in, "the route from one place to another")*
- (a) rhymes with "hoot"
 - (b) rhymes with "out"
 - (c) I can pronounce it either way interchangeably
 - (d) I say it like "hoot" for the noun and like "out" for the verb.
 - (e) I say it like "out" for the noun and like "hoot" for the verb.
 - (f) other
26. Question 267: *the first vowel in "syrup"*
- (a) i "sear-up"
 - (b) "sih-rup"
 - (c) as in "sir"
 - (d) other
27. Question 268: *Do you pronounce "cot" and "caught" the same?*
- (a) different
 - (b) same
28. Question 269: *almond*
- (a) all-mond (first syllable sounds like "all")
 - (b) ah-mond (no l)
 - (c) aw-mond (if different from "ah-mond")
 - (d) I say something in between l and nothing
 - (e) other
29. Question 270: *the "s" in "anniversary"*
- (a) s as in "sock"
 - (b) as in "shock"
30. Question 271: *asterisk*
- (a) asteriks

- (b) asterisk
 - (c) asterik (with no s in the final cluster)
 - (d) other
31. Question 272: *candidate*
- (a) I pronounce the first d
 - (b) I don't pronounce the first d
 - (c) I vary freely between pronouncing the first d and not doing so
 - (d) I only pronounce the first d when I'm speaking slowly/carefully
 - (e) Depends whether it refers to a political or generic candidate, as in "that assignment looks like a good candidate for elimination" (please state how the two pronunciations differ)
 - (f) other
32. Question 273: *the "s" in "chromosome"*
- (a) s
 - (b) z
 - (c) both are acceptable to me
 - (d) other
33. Question 274: *et cetera*
- (a) pronounced etsetera (4 syllables)
 - (b) pronounced etsetra (3 syllables)
 - (c) pronounced eksetera (4 syllables)
 - (d) pronounced eksetra (3 syllables)
 - (e) other
34. Question 275: *the final consonant in "garage"*
- (a) as in the middle consonant of "measure"
 - (b) as in "edge"
 - (c) I use both interchangeably
 - (d) other
35. Question 276: *the "c" in "grocery"*
- (a) s as in "sock"
 - (b) as in "shock"
 - (c) other
36. Question 277: *huge, humor, humongous, human...*
- (a) I pronounce the h
 - (b) I don't pronounce the h
 - (c) I can pronounce the h or not
 - (d) other
37. Question 278: *the "s" in "nursery"*
- (a) s as in "sock"
 - (b) as in "shock"
 - (c) other
38. Question 279: *the "s" in the last name of Elvis Presley*
- (a) s
 - (b) z
39. Question 280: *quarter*
- (a) with kw
 - (b) with k ("cor-ter")
 - (c) I use both interchangeably

- (d) other
- 40. Question 281: *Do you use "spigot" or "spicket" to refer to a faucet or tap that water comes out of?*
 - (a) spicket
 - (b) spigot
 - (c) I use both interchangeably
 - (d) I say "spicket" but spell it "spigot"
 - (e) I use both with different meanings (please explain how in the comments box)
 - (f) I don't use either version of this word
 - (g) other
- 41. Question 282: *strength*
 - (a) the "g" is pronounced as
 - (b) the "g" is pronounced as k
 - (c) the "g" is silent
- 42. Question 283: *the final consonant in "Texas"*
 - (a) s
 - (b) z
 - (c) either one
 - (d) other
- 43. Question 284: *cream cheese*
 - (a) CREAM cheese (stress on the first syllable)
 - (b) cream CHEESE (stress on the second syllable)
 - (c) it sounds right either way
 - (d) other
- 44. Question 285: *insurance*
 - (a) INsurance (stress on the first syllable)
 - (b) inSURance (stress on the second syllable)
 - (c) I can stress either the first or the second syllable
 - (d) other
- 45. Question 286: *New Haven (the city in Connecticut where Yale University is located)*
 - (a) NEW Haven
 - (b) New HAVEN
 - (c) I use both interchangeably
 - (d) other
- 46. Question 287: *Thanksgiving*
 - (a) THANKSgiving
 - (b) ThanksGIVing
 - (c) I use both interchangeably
 - (d) other
- 47. Question 288: *umbrella*
 - (a) UMBrella
 - (b) umBRELLa
- 48. Question 289: *I her lifeless body from the pool*
 - (a) dragged
 - (b) drug
 - (c) I use both interchangeably
 - (d) other

49. Question 291: *Would you say "Are you coming with?" as a full sentence, to mean "Are you coming with us?"*
- (a) yes
 - (b) no
 - (c) other
50. Question 292: *Would you say "where are you at?" to mean "where are you?"*
- (a) yes
 - (b) no
 - (c) I can use "where are you at" in contexts such as asking someone how s/he is coming along on a project, but not in the general sense of "where are you physically located in the world at this moment".
51. Question 293: *Modals are words like "can," "could," "might," "ought to," and so on. Can you use more than one modal at a time? (e.g., "I might could do that" to mean "I might be able to do that" or "I used to could do that" to mean "I used to be able to do that")*
- (a) yes (please consider adding which combinations of modals you use in the comments box)
 - (b) no
 - (c) other
52. Question 294: *He used to nap on the couch, but he sprawls out in that new lounge chair anymore*
- (a) this use of "anymore" is acceptable
 - (b) this use of "anymore" is unacceptable
 - (c) not sure
53. Question 295: *I do exclusively figurative paintings anymore*
- (a) acceptable
 - (b) unacceptable
 - (c) not sure
54. Question 296: *Panty hose are so expensive anymore that I just try to get a good suntan and forget about it.*
- (a) acceptable
 - (b) unacceptable
 - (c) not sure
55. Question 297: *Forget the nice clothes anymore (referring to babies eating messily after a certain age)*
- (a) acceptable
 - (b) unacceptable
 - (c) not sure
56. Question 299: *What do you call the game wherein the participants see who can throw a knife closest to the other person (or alternately, get a jackknife to stick into the ground or a piece of wood)?*
- (a) mumblety-peg
 - (b) mumbledy-peg
 - (c) mumbly peg
 - (d) mumbly pegs
 - (e) mumblely peg (with 2 l's)
 - (f) mumble peg
 - (g) mummety-peg
 - (h) mumble-the-peg
 - (i) fumbledy peg
 - (j) numblety peg

- (k) peggy
 - (l) baseball jackknife
 - (m) stick-knife
 - (n) stick-frog
 - (o) stretch
 - (p) chicken
 - (q) knifey
 - (r) splits
 - (s) Russian roulette
 - (t) I have never heard of this "game" and have no idea what it's called
 - (u) other (state here if you have heard one or more of these terms but never knew what they meant)
57. Question 300: *What do you call the area of grass between the sidewalk and the road?*
- (a) berm
 - (b) parking
 - (c) tree lawn
 - (d) terrace
 - (e) curb strip
 - (f) beltway
 - (g) verge
 - (h) I have no word for this
 - (i) other
58. Question 301: *What do you call the area of grass that occurs in the middle of some streets?*
- (a) boulevard
 - (b) midway
 - (c) traffic island
 - (d) island
 - (e) neutral ground
 - (f) I have no word for this
 - (g) other
59. Question 302: *What do you call the long narrow place in the middle of a divided highway?*
- (a) median strip
 - (b) median
 - (c) boulevard
 - (d) mall
 - (e) traffic island
 - (f) neutral ground
 - (g) island
 - (h) park strip
 - (i) I have no word for this
 - (j) other
60. Question 303: *What do you call the drink made with milk and ice cream?*
- (a) milkshake/shake
 - (b) frappe
 - (c) cabinet
 - (d) velvet
 - (e) thick shake
 - (f) other
61. Question 305: *What do you call the insect that flies around in the summer and has a rear section that glows in the dark?*

- (a) lightning bug
 - (b) firefly
 - (c) I use lightning bug and firefly interchangeably
 - (d) peenie wallie
 - (e) I have no word for this
 - (f) other
62. Question 307: *What do you call the kind of spider (or spider-like creature) that has an oval-shaped body and extremely long legs?*
- (a) daddy long leg(s)
 - (b) daddy big legs
 - (c) daddy (bug)
 - (d) father longlegs
 - (e) granddaddy
 - (f) daddy graybeard
 - (g) daddy spider
 - (h) harvestman
 - (i) moskeet spider
 - (j) pointer
 - (k) shepherd spider
 - (l) other
63. Question 308: *What nicknames do/did you use for your maternal grandmother?*
- (a) grandmother
 - (b) granny
 - (c) grandma
 - (d) nana
 - (e) mimi
 - (f) grammy/grammie/grammi
 - (g) other
64. Question 309: *What about your paternal grandmother (is there a distinction?)*
- (a) grandmother
 - (b) granny
 - (c) grandma
 - (d) gramma
 - (e) nana
 - (f) other
65. Question 310: *What do/did you call your maternal grandfather?*
- (a) gramps
 - (b) grandpa
 - (c) grampa
 - (d) grandad, granddad
 - (e) pap
 - (f) I spell it "grandpa" but pronounce it as "grampa"
 - (g) other (including if you use a different term to address him directly than you do when speaking about him to a third party)
66. Question 311: *paternal grandfather?*
- (a) gramps
 - (b) grandpa
 - (c) grampa
 - (d) pap

- (e) other
67. Question 312: *What do you call the big clumps of dust that gather under furniture and in corners?*
- (a) dust bunnies
 - (b) dust kittens
 - (c) dust mice
 - (d) kitties
 - (e) dust balls
 - (f) other
68. Question 316: *What term do you use to refer to something that is across both streets from you at an intersection (or diagonally across from you in general)?*
- (a) kitty-corner
 - (b) kitacorn
 - (c) catercorner
 - (d) catty-corner
 - (e) kitty cross
 - (f) kitty wampus
 - (g) I can only use "diagonal" for this
 - (h) I have no term for this
 - (i) other
69. Question 317: *What do you call the activity of driving around in circles in a car?*
- (a) doing donuts
 - (b) doing cookies
 - (c) whipping shitties
 - (d) other
70. Question 318: *What do you call paper that has already been used for something or is otherwise imperfect?*
- (a) scratch paper
 - (b) scrap paper
 - (c) scratch paper is still usable (for example, the paper you bring to do extra work on a test) scrap paper is paper that isn't needed anymore and can be thrown away.
 - (d) other
71. Question 319: *What is your general term for a big road that you drive relatively fast on?*
- (a) highway
 - (b) freeway
 - (c) parkway
 - (d) turnpike
 - (e) expressway
 - (f) throughway/thru-way
 - (g) a freeway is bigger than a highway
 - (h) a freeway is free (i.e., doesn't charge tolls) a highway isn't
 - (i) a freeway has limited access (no stop lights, no intersections), whereas a highway can have stop lights and intersections
 - (j) other
72. Question 321: *When you are cold, and little points of skin begin to come on your arms and legs, you have-*
- (a) goose bumps
 - (b) goose flesh
 - (c) goose pimples
 - (d) chill bumps

- (e) chill bugs
 - (f) chilly bumps
 - (g) cold-chill bumps
 - (h) other
73. Question 323: *What do you call an easy course?*
- (a) gut
 - (b) crypt course
 - (c) crip course
 - (d) bird
 - (e) blow-off
 - (f) meat
 - (g) other
74. Question 325: *What is the thing that women use to tie their hair?*
- (a) (hair) elastic
 - (b) rubber band
 - (c) horsetail
 - (d) hair thing
 - (e) hair tie
 - (f) other
75. Question 326: *Do you use the word cruller?*
- (a) yes
 - (b) no, but I know what it means
 - (c) I have no idea what this means
76. Question 327: *Do you use the term "bear claw" for a kind of pastry?*
- (a) yes
 - (b) no, but I know what it means
 - (c) I have no idea what this means
77. Question 328: *What do you call someone who is the opposite of pigeon-toed (i.e. when they walk their feet point outwards)?*
- (a) duck-footed
 - (b) slue-footed
 - (c) splay-footed
 - (d) bow-legged
 - (e) toed out
 - (f) other
 - (g) I have no word for this
78. Question 329: *Can you call coleslaw "slaw"?*
- (a) yes
 - (b) yes, but I can also use it in other forms such as apple slaw or broccoli slaw
 - (c) no
 - (d) I have never heard that usage before
 - (e) other
79. Question 330: *What do you call the box you bury a dead person in?*
- (a) coffin
 - (b) casket
 - (c) a coffin and a casket are not the same, and I know the difference
 - (d) other
80. Question 331: *Do you say "vinegar and oil" or "oil and vinegar" for the type of salad dressing?*

- (a) vinegar and oil
 - (b) oil and vinegar
 - (c) both sound equally good to me
 - (d) neither
 - (e) other
81. Question 332: *What do you call it when a driver changes over one or more lanes way too quickly?*
- (a) Chinese lane change
 - (b) Chinese fire drill
 - (c) other
82. Question 333: *When you stand outside with a long line of people waiting to get in somewhere, are you standing "in line" or "on line" (as in, "I stood in the cold for two hours before they opened the doors")?*
- (a) on line
 - (b) in line
 - (c) both sound equally good
 - (d) neither
 - (e) other
83. Question 334: *Do you say "frosting" or "icing" for the sweet spread one puts on a cake?*
- (a) frosting
 - (b) icing
 - (c) icing is thinner than frosting, white, and/or made of powdered sugar and milk or lemon juice
 - (d) both
 - (e) neither
 - (f) other
84. Question 335: *What is "the City"?*
- (a) New York City
 - (b) Boston
 - (c) DC
 - (d) LA
 - (e) Chicago
 - (f) other
85. Question 336: *What is the distinction between dinner and supper?*
- (a) supper is an evening meal while dinner is eaten earlier (lunch, for example)
 - (b) supper is an evening meal, dinner is the main meal
 - (c) dinner takes place in a more formal setting than supper
 - (d) there is no distinction they both have the same meaning
 - (e) I do not use the term supper
 - (f) I don't use the term dinner
 - (g) other
86. Question 337: *Which of these terms do you prefer?*
- (a) trash can
 - (b) garbage can
 - (c) rubbish bin
 - (d) waste(paper) basket
 - (e) These words refer to different things
 - (f) other
87. Question 338: *Which of these terms do you prefer?*

- (a) By accident
 - (b) On accident
 - (c) both
 - (d) neither
 - (e) other
88. Question 339: *Which of these terms do you prefer for the small road parallel to the highway?*
- (a) frontage road
 - (b) service road
 - (c) access road
 - (d) feeder road
 - (e) gateway
 - (f) we have them but I have no word for them
 - (g) I've never heard of this concept
 - (h) other
89. Question 340: *Do you cut or mow the lawn or grass?*
- (a) cut the grass
 - (b) cut the lawn
 - (c) mow the grass
 - (d) mow the lawn
 - (e) other
90. Question 341: *Do you pass in homework or hand in homework?*
- (a) pass in
 - (b) hand in
 - (c) both
 - (d) neither
 - (e) other
91. Question 343: *What do you call the thing from which you might drink water in a school?*
- (a) bubbler
 - (b) water bubbler
 - (c) drinking fountain
 - (d) water fountain
 - (e) other
92. Question 344: *What do you call a public railway system (normally underground)?*
- (a) the subway
 - (b) the L, or the El
 - (c) the T
 - (d) the metro
 - (e) BART
 - (f) other
93. Question 346: *What do you call the act of covering a house or area in front of a house with toilet paper?*
- (a) tp'ing
 - (b) rolling
 - (c) toilet papering
 - (d) wrapping
 - (e) papering
 - (f) bog rolling
 - (g) I have no word for this
 - (h) other

94. Question 347: *What do you call a traffic jam caused by drivers slowing down to look at an accident or other diversion on the side of the road?*
- (a) rubberneck
 - (b) rubbernecking
 - (c) rubbernecking is the activity (slowing down and gawking) that causes the traffic jam, but I have no word for the traffic jam itself
 - (d) gapers' block
 - (e) gapers' delay
 - (f) Lookie Lou
 - (g) curiosity delay
 - (h) gawk block
 - (i) I have no word for this
 - (j) other
95. Question 348: *What vowel do you use in bag?*
- (a) as in "sat"
 - (b) as in "set"
 - (c) e as in "say"
 - (d) other
96. Question 349: *What do you call the paper container in which you might bring home items you bought at the store?*
- (a) bag
 - (b) sack
 - (c) poke
 - (d) other
97. Question 350: *What do you call the night before Halloween?*
- (a) gate night
 - (b) trick night
 - (c) mischief night
 - (d) cabbage night
 - (e) goosy night
 - (f) devil's night
 - (g) devil's eve
 - (h) I have no word for this
 - (i) other
98. Question 351: *What do you call the end of a loaf of bread?*
- (a) end
 - (b) heel
 - (c) crust
 - (d) nose
 - (e) butt
 - (f) shpitzel
 - (g) I have no word for this
 - (h) other
99. Question 352: *How do you pronounce the word for the type of drug that acts as central nervous system depressant and is used as a sedative or hypnotic? (Please do not look up the word in a dictionary before answering this question.)*
- (a) barbituate
 - (b) barbiturate
 - (c) I don't use either of these
 - (d) other

100. Question 353: *amphitheater*
- (a) f
 - (b) p
 - (c) other
101. Question 354: *citizen*
- (a) s
 - (b) z
 - (c) other
102. Question 355: *What do you call a point that is purely academic, or that cannot be settled and isn't worth discussing further?*
- (a) a moot point
 - (b) a mute point
 - (c) either one of the above
 - (d) I have no idea
 - (e) other
103. Question 356: *How do you pronounce the -sp- sequence in "thespian" (the word meaning "actor")?*
- (a) sp (as in "desperate")
 - (b) zb (rhymes with "lesbian")
 - (c) other
104. Question 357: *What do you call the level of a building that is partly or entirely underground?*
- (a) basement
 - (b) cellar
 - (c) I use both, and they mean the same thing
 - (d) A basement is finished (for example with plastered or painted walls, carpets, etc.), whereas a cellar is unfinished (made up of bare stone or cement, used only for storage).
 - (e) A cellar has an outside entrance (some call this a "bulkhead"), whereas a basement does not
 - (f) other
105. Question 358: *What do you call a drive-through liquor store?*
- (a) brew thru
 - (b) party barn
 - (c) bootlegger
 - (d) beer barn
 - (e) beverage barn
 - (f) we have these in my area, but we have no special term for them
 - (g) I have never heard of such a thing
 - (h) other
106. Question 360: *What do you say when you want to lay claim to the front seat of a car?*
- (a) dibs
 - (b) shotgun
 - (c) hosey
 - (d) high hosey
 - (e) I have no term for this
 - (f) other
107. Question 361: *What word do you use for gawking at someone in a lustful way?*
- (a) ogle
 - (b) oogle
 - (c) oggle (pronounced to rhyme with "boggle", but may still be spelled "ogle")

- (d) I use both oogle and ogle interchangeably
 - (e) I use both ogle and "oggle"
 - (f) I have no word for this activity
 - (g) other
108. Question 362: *Do you say "expecially", or "especially"?*
- (a) expecially (or "ecpecially" or "ekpecially")
 - (b) especially
 - (c) I use both interchangeably
 - (d) neither
 - (e) other

Appendix B: Top 50 Survey Questions Identified by Logistic Regression Feature Coefficients

1. Question 245: *the vowel in the second syllable of "cauliflower"*
 - (a) i as in "see"
 - (b) as in "sit"
 - (c) other
2. Question 249: *crayon*
 - (a) as in "man" (1 syllable, "cran")
 - (b) ej (2 syllables, "cray-ahn")
 - (c) ej (2 syllables, "cray-awn", where the second syllable rhymes with "dawn")
 - (d) aw (I pronounce this the same as "crown")
 - (e) other
3. Question 250: *creek (a small body of running water)*
 - (a) i as in "see"
 - (b) as in "sit"
 - (c) I use both interchangeably
 - (d) I don't know how to pronounce this word
 - (e) I use both, but they mean two different things (please state how they differ in the comments box)
 - (f) other
4. Question 253: *the last vowel in "handkerchief"*
 - (a) i as in "see"
 - (b) as in "sit"
 - (c) other
5. Question 255: *How do you pronounce Mary/merry/marry?*
 - (a) all 3 are the same
 - (b) all 3 are different
 - (c) Mary and merry are the same marry is different
 - (d) merry and marry are the same Mary is different
 - (e) Mary and marry are the same merry is different
6. Question 267: *the first vowel in "syrup"*
 - (a) i "sear-up"
 - (b) "sih-rup"
 - (c) as in "sir"
 - (d) other
7. Question 268: *Do you pronounce "cot" and "caught" the same?*
 - (a) different

- (b) same
- 8. Question 272: *candidate*
 - (a) I pronounce the first d
 - (b) I don't pronounce the first d
 - (c) I vary freely between pronouncing the first d and not doing so
 - (d) I only pronounce the first d when I'm speaking slowly/carefully
 - (e) Depends whether it refers to a political or generic candidate, as in "that assignment looks like a good candidate for elimination" (please state how the two pronunciations differ)
 - (f) other
- 9. Question 277: *huge, humor, humongous, human...*
 - (a) I pronounce the h
 - (b) I don't pronounce the h
 - (c) I can pronounce the h or not
 - (d) other
- 10. Question 278: *the "s" in "nursery"*
 - (a) s as in "sock"
 - (b) as in "shock"
 - (c) other
- 11. Question 281: *Do you use "spigot" or "spicket" to refer to a faucet or tap that water comes out of?*
 - (a) spicket
 - (b) spigot
 - (c) I use both interchangeably
 - (d) I say "spicket" but spell it "spigot"
 - (e) I use both with different meanings (please explain how in the comments box)
 - (f) I don't use either version of this word
 - (g) other
- 12. Question 283: *the final consonant in "Texas"*
 - (a) s
 - (b) z
 - (c) either one
 - (d) other
- 13. Question 285: *insurance*
 - (a) INsurance (stress on the first syllable)
 - (b) inSURance (stress on the second syllable)
 - (c) I can stress either the first or the second syllable
 - (d) other
- 14. Question 286: *New Haven (the city in Connecticut where Yale University is located)*
 - (a) NEW Haven
 - (b) New HAVEN
 - (c) I use both interchangeably
 - (d) other
- 15. Question 299: *What do you call the game wherein the participants see who can throw a knife closest to the other person (or alternately, get a jackknife to stick into the ground or a piece of wood)?*
 - (a) mumblety-peg
 - (b) mumbledy-peg
 - (c) mumbly peg

- (d) mumbly pegs
 - (e) mumblely peg (with 2 l's)
 - (f) mumble peg
 - (g) mummety-peg
 - (h) mumble-the-peg
 - (i) fumbledy peg
 - (j) numblety peg
 - (k) peggy
 - (l) baseball jackknife
 - (m) stick-knife
 - (n) stick-frog
 - (o) stretch
 - (p) chicken
 - (q) knifey
 - (r) splits
 - (s) Russian roulette
 - (t) I have never heard of this "game" and have no idea what it's called
 - (u) other (state here if you have heard one or more of these terms but never knew what they meant)
16. Question 300: *What do you call the area of grass between the sidewalk and the road?*
- (a) berm
 - (b) parking
 - (c) tree lawn
 - (d) terrace
 - (e) curb strip
 - (f) beltway
 - (g) verge
 - (h) I have no word for this
 - (i) other
17. Question 301: *What do you call the area of grass that occurs in the middle of some streets?*
- (a) boulevard
 - (b) midway
 - (c) traffic island
 - (d) island
 - (e) neutral ground
 - (f) I have no word for this
 - (g) other
18. Question 302: *What do you call the long narrow place in the middle of a divided highway?*
- (a) median strip
 - (b) median
 - (c) boulevard
 - (d) mall
 - (e) traffic island
 - (f) neutral ground
 - (g) island
 - (h) park strip
 - (i) I have no word for this
 - (j) other
19. Question 303: *What do you call the drink made with milk and ice cream?*
- (a) milkshake/shake

- (b) frappe
 - (c) cabinet
 - (d) velvet
 - (e) thick shake
 - (f) other
20. Question 305: *What do you call the insect that flies around in the summer and has a rear section that glows in the dark?*
- (a) lightning bug
 - (b) firefly
 - (c) I use lightning bug and firefly interchangeably
 - (d) peenie wallie
 - (e) I have no word for this
 - (f) other
21. Question 307: *What do you call the kind of spider (or spider-like creature) that has an oval-shaped body and extremely long legs?*
- (a) daddy long leg(s)
 - (b) daddy big legs
 - (c) daddy (bug)
 - (d) father longlegs
 - (e) granddaddy
 - (f) daddy graybeard
 - (g) daddy spider
 - (h) harvestman
 - (i) moskeet spider
 - (j) pointer
 - (k) shepherd spider
 - (l) other
22. Question 310: *What do/did you call your maternal grandfather?*
- (a) gramps
 - (b) grandpa
 - (c) grampa
 - (d) grandad, granddad
 - (e) pap
 - (f) I spell it "grandpa" but pronounce it as "grampa"
 - (g) other (including if you use a different term to address him directly than you do when speaking about him to a third party)
23. Question 312: *What do you call the big clumps of dust that gather under furniture and in corners?*
- (a) dust bunnies
 - (b) dust kittens
 - (c) dust mice
 - (d) kitties
 - (e) dust balls
 - (f) other
24. Question 316: *What term do you use to refer to something that is across both streets from you at an intersection (or diagonally across from you in general)?*
- (a) kitty-corner
 - (b) kitacornor
 - (c) catercorner
 - (d) catty-corner

- (e) kitty cross
 - (f) kitty wampus
 - (g) I can only use "diagonal" for this
 - (h) I have no term for this
 - (i) other
25. Question 317: *What do you call the activity of driving around in circles in a car?*
- (a) doing donuts
 - (b) doing cookies
 - (c) whipping shitties
 - (d) other
26. Question 319: *What is your general term for a big road that you drive relatively fast on?*
- (a) highway
 - (b) freeway
 - (c) parkway
 - (d) turnpike
 - (e) expressway
 - (f) throughway/thru-way
 - (g) a freeway is bigger than a highway
 - (h) a freeway is free (i.e., doesn't charge tolls) a highway isn't
 - (i) a freeway has limited access (no stop lights, no intersections), whereas a highway can have stop lights and intersections
 - (j) other
27. Question 321: *When you are cold, and little points of skin begin to come on your arms and legs, you have-*
- (a) goose bumps
 - (b) goose flesh
 - (c) goose pimples
 - (d) chill bumps
 - (e) chill bugs
 - (f) chilly bumps
 - (g) cold-chill bumps
 - (h) other
28. Question 323: *What do you call an easy course?*
- (a) gut
 - (b) crypt course
 - (c) crip course
 - (d) bird
 - (e) blow-off
 - (f) meat
 - (g) other
29. Question 325: *What is the thing that women use to tie their hair?*
- (a) (hair) elastic
 - (b) rubber band
 - (c) horsetail
 - (d) hair thing
 - (e) hair tie
 - (f) other
30. Question 333: *When you stand outside with a long line of people waiting to get in somewhere, are you standing "in line" or "on line" (as in, "I stood in the cold for two hours before they opened the doors")?*

- (a) on line
 - (b) in line
 - (c) both sound equally good
 - (d) neither
 - (e) other
31. Question 334: *Do you say "frosting" or "icing" for the sweet spread one puts on a cake?*
- (a) frosting
 - (b) icing
 - (c) icing is thinner than frosting, white, and/or made of powdered sugar and milk or lemon juice
 - (d) both
 - (e) neither
 - (f) other
32. Question 335: *What is "the City"?*
- (a) New York City
 - (b) Boston
 - (c) DC
 - (d) LA
 - (e) Chicago
 - (f) other
33. Question 336: *What is the distinction between dinner and supper?*
- (a) supper is an evening meal while dinner is eaten earlier (lunch, for example)
 - (b) supper is an evening meal, dinner is the main meal
 - (c) dinner takes place in a more formal setting than supper
 - (d) there is no distinction they both have the same meaning
 - (e) I do not use the term supper
 - (f) I don't use the term dinner
 - (g) other
34. Question 337: *Which of these terms do you prefer?*
- (a) trash can
 - (b) garbage can
 - (c) rubbish bin
 - (d) waste(paper) basket
 - (e) These words refer to different things
 - (f) other
35. Question 338: *Which of these terms do you prefer?*
- (a) By accident
 - (b) On accident
 - (c) both
 - (d) neither
 - (e) other
36. Question 339: *Which of these terms do you prefer for the small road parallel to the highway?*
- (a) frontage road
 - (b) service road
 - (c) access road
 - (d) feeder road
 - (e) gateway
 - (f) we have them but I have no word for them
 - (g) I've never heard of this concept

- (h) other
- 37. Question 341: *Do you pass in homework or hand in homework?*
 - (a) pass in
 - (b) hand in
 - (c) both
 - (d) neither
 - (e) other
- 38. Question 343: *What do you call the thing from which you might drink water in a school?*
 - (a) bubbler
 - (b) water bubbler
 - (c) drinking fountain
 - (d) water fountain
 - (e) other
- 39. Question 344: *What do you call a public railway system (normally underground)?*
 - (a) the subway
 - (b) the L, or the El
 - (c) the T
 - (d) the metro
 - (e) BART
 - (f) other
- 40. Question 346: *What do you call the act of covering a house or area in front of a house with toilet paper?*
 - (a) tp'ing
 - (b) rolling
 - (c) toilet papering
 - (d) wrapping
 - (e) papering
 - (f) bog rolling
 - (g) I have no word for this
 - (h) other
- 41. Question 347: *What do you call a traffic jam caused by drivers slowing down to look at an accident or other diversion on the side of the road?*
 - (a) rubberneck
 - (b) rubbernecking
 - (c) rubbernecking is the activity (slowing down and gawking) that causes the traffic jam, but I have no word for the traffic jam itself
 - (d) gapers' block
 - (e) gapers' delay
 - (f) Lookie Lou
 - (g) curiosity delay
 - (h) gawk block
 - (i) I have no word for this
 - (j) other
- 42. Question 349: *What do you call the paper container in which you might bring home items you bought at the store?*
 - (a) bag
 - (b) sack
 - (c) poke
 - (d) other

43. Question 350: *What do you call the night before Halloween?*
- (a) gate night
 - (b) trick night
 - (c) mischief night
 - (d) cabbage night
 - (e) goosy night
 - (f) devil's night
 - (g) devil's eve
 - (h) I have no word for this
 - (i) other
44. Question 351: *What do you call the end of a loaf of bread?*
- (a) end
 - (b) heel
 - (c) crust
 - (d) nose
 - (e) butt
 - (f) shpitzel
 - (g) I have no word for this
 - (h) other
45. Question 352: *How do you pronounce the word for the type of drug that acts as central nervous system depressant and is used as a sedative or hypnotic? (Please do not look up the word in a dictionary before answering this question.)*
- (a) barbituate
 - (b) barbiturate
 - (c) I don't use either of these
 - (d) other
46. Question 355: *What do you call a point that is purely academic, or that cannot be settled and isn't worth discussing further?*
- (a) a moot point
 - (b) a mute point
 - (c) either one of the above
 - (d) I have no idea
 - (e) other
47. Question 356: *How do you pronounce the -sp- sequence in "thespian" (the word meaning "actor")?*
- (a) sp (as in "desperate")
 - (b) zb (rhymes with "lesbian")
 - (c) other
48. Question 357: *What do you call the level of a building that is partly or entirely underground?*
- (a) basement
 - (b) cellar
 - (c) I use both, and they mean the same thing
 - (d) A basement is finished (for example with plastered or painted walls, carpets, etc.), whereas a cellar is unfinished (made up of bare stone or cement, used only for storage).
 - (e) A cellar has an outside entrance (some call this a "bulkhead"), whereas a basement does not
 - (f) other
49. Question 358: *What do you call a drive-through liquor store?*
- (a) brew thru

- (b) party barn
- (c) bootlegger
- (d) beer barn
- (e) beverage barn
- (f) we have these in my area, but we have no special term for them
- (g) I have never heard of such a thing
- (h) other

50. Question 360: *What do you say when you want to lay claim to the front seat of a car?*

- (a) dibs
- (b) shotgun
- (c) hosey
- (d) high hosey
- (e) I have no term for this
- (f) other

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References

John Leverso, Youness Diouane, and George Mohler. Measuring online–offline spillover of gang violence using bivariate hawkes processes. *Journal of quantitative criminology*, 41(1):103–131, 2025.